Rongge Xu

Higher Category Theory · Mathematical Physics · AI for Mathematics

rongge@tsinghua.edu.cn | github.com/ronggexu | (+86) 13651822330



Research Profile

My research focuses on higher category theory and its applications to mathematical physics, particularly higher Morita theory and topological orders. I also work on Lean-based formalization and AI for mathematics, with a broader interest in the structural organization of mathematical knowledge.

Academic Positions

Yau Mathematical Sciences Center, Tsinghua University — *Beijing, China*

Postdoctoral Fellow — *2025 – present*

- Research areas: higher category theory, higher Morita theory, topological orders, and AI for mathematics.

Education

Fudan University; joint training at Westlake University — *Shanghai/Hangzhou, China*

Ph.D. in Physics — *2018 – 2025*

- Advisors: Prof. Jian Li (condensed matter physics) and Prof. Liang Kong (category theory and topological order).
- Research: categorical descriptions of topological phases.

University of Minnesota, Twin Cities — *Minnesota, USA*

B.S. in Architecture — *2013 – 2017*

- Minor in Landscape Planning. Exchange program in Scandinavia (Copenhagen, 2016–2017).

Publications

Research Articles and Preprints

- **Rongge Xu**, Holiverse Yang. *2-Morita Theory of E_2-Algebras and Module Categories*. arXiv:2608.27228, 2026.
- **Rongge Xu**, Hui Dai, Yiming Fu, Jiedong Jiang, Tianjiao Nie, Junkai Wang, Holiverse Yang, Zhi-Hao Zhang. *LeanCat: A Benchmark Suite for Formal Category Theory in Lean (Part I: 1-Categories)*. arXiv:2512.24796, 2025 (revised 2026); accepted at the 3rd AI for Math Workshop, ICML 2026. sciencraft/LeanCat.
- **Rongge Xu**, Holiverse Yang. *Classification of Gapped Domain Walls in 2+1D Topological Orders through 2-Morita Equivalence*. Chinese Physics Letters, 42(7), 070710, 2025.
- **Rongge Xu**, Zhi-Hao Zhang. *Categorical descriptions of one-dimensional gapped phases with Abelian onsite symmetries*. Physical Review B, 110(15), 155106, 2024.
- L. Guo, Y. Yan, **R. Xu**, J. Li, C. Zeng. *Zero-Bias Conductance Peaks Effectively Tuned by Gating-Controlled Rashba Spin-Orbit Coupling*. Physical Review Letters, 126(5), 057701, 2021.

Survey Articles

- Rongge Xu. *Recent Advances in Formal Mathematics with Lean*. ICCM Notices, forthcoming.

Selected Research Projects

- **LeanCat** — A Lean 4/Mathlib benchmark of 100 category-theory problems for evaluating library-grounded, interface-level reasoning by AI proving agents. sciencraft/LeanCat.
- **LeanBridge** — A retrieval-augmented agentic proving system for Lean, combining library search, generation, compiler feedback, and proof repair. sciencraft/leanbridge.
- **Formalization of Mathematical Physics in Lean** — Ongoing project on formalizing arguments in special relativity and electromagnetism, with emphasis on hidden assumptions, axiom choice, and machine-checkable physical reasoning.

Selected Talks and Presentations

- *Aug. 30, 2026* *2-Morita Theory of E_2-Algebras and Module-Categories*. Postdoctoral Presentation, YMSC, Tsinghua University.
- *Jul. 23, 2026* *Lean and AI for Physics: Formalization, Semantics, and Discovery*. Seminar, Westlake University.
- *Jul. 2026* *LeanCat: A Benchmark Suite for Formal Category Theory in Lean (Part I: 1-Categories)*. Poster, 3rd AI for Math Workshop, ICML 2026, Seoul.
- *Jun. 2026* *Formalization as a Tool for Mathematical Physics: Categorical Benchmark and Special Relativity in Lean*. Postdoctoral Presentation, YMSC, Tsinghua University.
- *Nov. 12, 2025* *Lean-based AI for Math: From Formal Proofs to Verifiable Discovery*. Seminar, Ubiquant AI for Math, Beijing.
- *Sep. 2024* *2-Morita Equivalent Condensable Algebras and Domain Walls in 2+1D Topological Orders*. SUSTech–Nagoya Workshop.
- *Jul. 2024* Poster & Gong Show, PiTP: *Ultra-Quantum Matter*, Institute for Advanced Study, Princeton.
- *Jun. 2024* Poster & Gong Show, IHES Summer School: *Symmetries and Anomalies*, France.

Teaching

- *Winter 2026* **Instructor**, Lean 4 Workshop, Qiuzhen College, Tsinghua University. Designed and taught a workshop on Lean 4 and Mathlib, including lectures, exercises, and assignments. The course was associated with YMSC's initiative to formalize the classification of finite simple groups. ronggexu/lean4_workshop_26winter.
- *Mar. 2023* **Lecturer**, Workshop on Category and Topological Order, Shenzhen.
- *Jul. 2022* **Lecturer**, Introduction to Category Theory in Physics, Westlake University.

Honors

- *Aug. 2025* **First Prize**, AI for Science Hackathon · Beijing, Tsinghua University (Project: LeanBridge)
- *2022* **National Scholarship**, Fudan University (Doctoral)
- *2022* **Outstanding Student Award**, Fudan University
- *2022–2023* Selected earlier awards: MindSpore Quantum Computing Hackathon Second Prize; Huawei Quantum Computing Hackathon Excellence Award; Ascend AI Innovation Competition Bronze Award.

Technical Skills

- **Formal Verification**: Lean 4, Mathlib, theorem proving, proof engineering, formalization of mathematics and physics.
- **AI for Mathematics**: LLM agents, retrieval-augmented theorem proving, compiler-feedback loops, benchmark design.
- Additional tools: Python, Julia, MATLAB, LaTeX, Git/GitHub; Illustrator, InDesign, Photoshop, Rhino, AutoCAD, Revit, SketchUp.

Patent

- *Apr. 2024* A Design Method and System for an Open Educational Game Driven by Generative Large Language Models. CN117899454A.